

UNIVERSITY OF ESWATINI

Department of Computer Science · CSC 400 / 402 / 499

How the supervision and assessment system works

A process handbook for supervisors, assessors, moderators and the project coordinator. Every flow in this document is the flow the software actually enforces, including the places where it refuses.

UPSAS — Research Project Supervision and Assessment System

Process flows, decision points and responsibilities

Generated from `docs/process-flow/build.py`

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1. The system at a glance

The system carries one undergraduate research project from a topic on a supervisor's list to a released mark on a student's record, and keeps enough of the reasoning to defend that mark a year later.

Three things make it more than a spreadsheet. Marks are normalised against the form they were awarded under, so a score out of 40 and a score out of 80 can be compared without anyone doing arithmetic in their head. Authority is checked per action rather than per person, which matters in a department where the same lecturer supervises, assesses and moderates. And a computed mark is stored as a snapshot, so the number a student sees and the number on a schedule cannot drift apart.

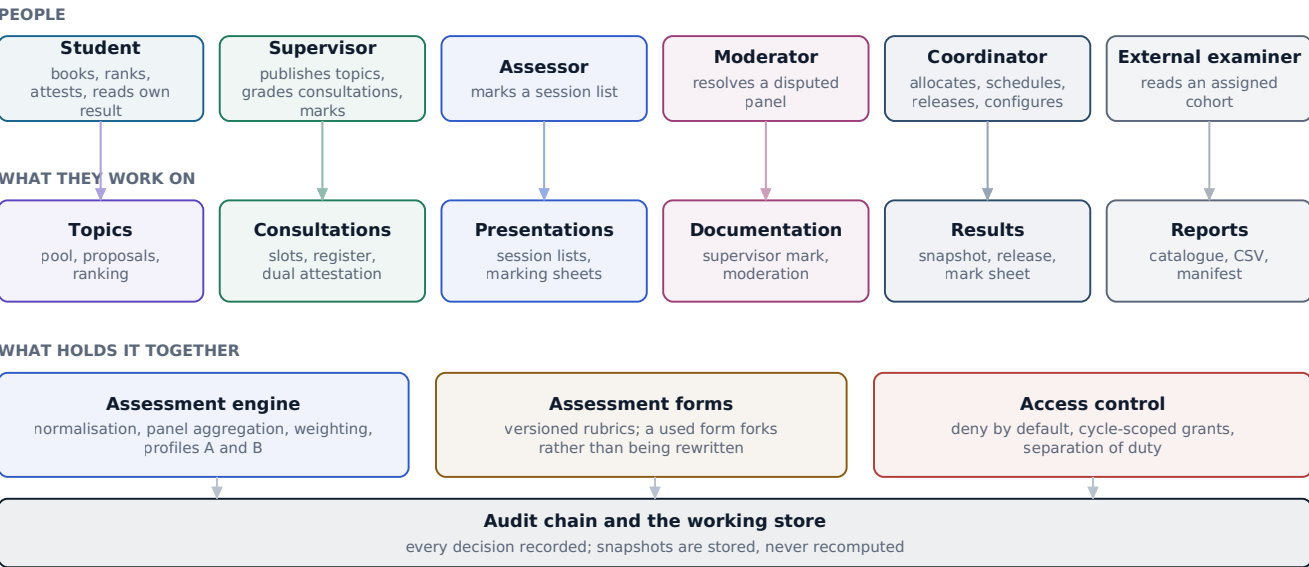


Figure 1. The parts of the system, and who touches which of them.

The rest of this document walks each flow in turn. Diamonds mark the points where the system refuses something; those are the parts worth reading closely, because they are where a process is protected rather than merely recorded.

2. Who does what

Roles are granted for a cycle and resolved on every request, so a grant withdrawn this morning is gone from the next page load rather than the next sign-in.

Role	What it may do
Student	Rank topics, propose one, book sessions, attest their own, read their own result
Supervisor	Publish topics, accept proposals, open slots, grade and attest consultations, nominate for presentation, mark documentation, assess their supervisees
Assessor	Mark the students on their scheduled session list, and nobody else
Moderator	Resolve a panel whose spread exceeds the threshold, with a written rationale
Coordinator	Allocate, schedule, reopen a sheet, moderate, release results, maintain the assessment forms, the cohort register and the deadline schedule, approve staff accounts
External examiner	Read an assigned cohort. The grant carries an expiry and closes itself
Administrator	Accounts and system configuration. The only role that may be granted outside a cycle

The hard case is not a person with one role. It is a lecturer holding four at once, which is ordinary in a department this size. Holding several roles unions their permissions and nothing more; separation of duty is then applied per action, against the person who actually produced the mark in question.

ONE LECTURER, SEVERAL HATS

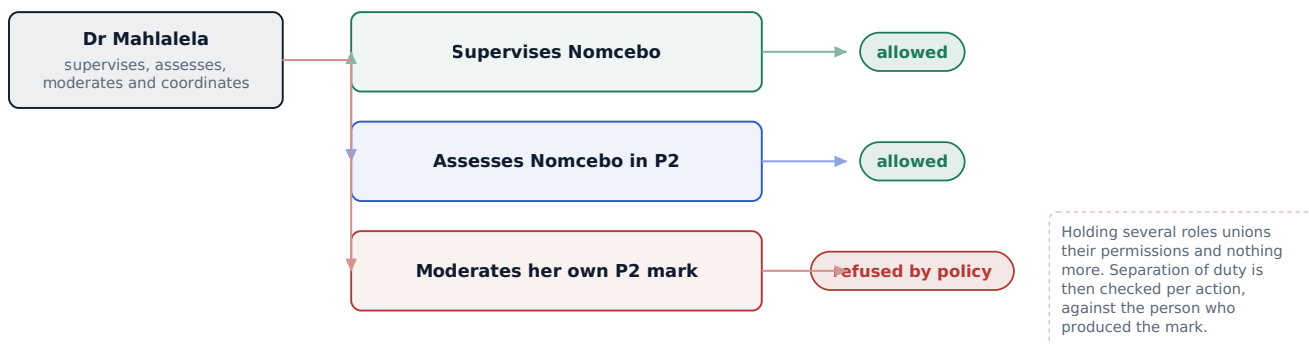


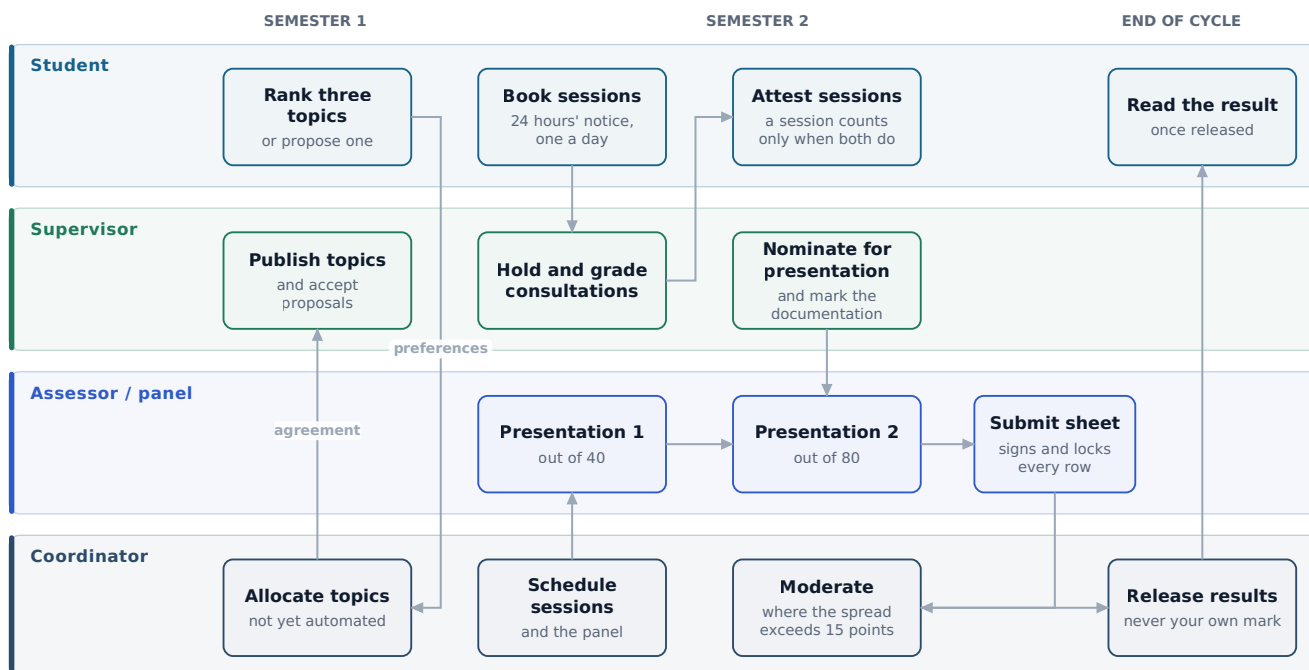
Figure 2. Roles accumulate; authority over your own work does not. The same check governs unlocking a sheet and releasing a result.

Why this is not just a role check

A rule that said "moderators may moderate" would let the lecturer who marked a presentation moderate their own mark, because they genuinely are a moderator. The check has to ask who produced the mark, not who is asking.

3. A cycle end to end

Consultations run all year. Everything else happens once, in this order.



Consultations run the whole year. Everything else happens once.

Figure 3. One cycle, by who is acting. The horizontal axis is time, roughly.

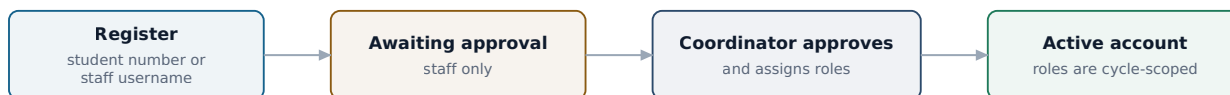
Two features of the shape are worth naming. Students may work individually or in pairs, and the system treats a pair as one project with two members: the proposal, artefact and documentation belong to the project, while consultations, contribution statements and marks belong to the student. Members of a pair can be, and often are, scored differently.

Second, a session number is not a team. The paper form groups two students per slot, which can mean one joint project or two unrelated students sharing a time. The grading sheet bands each session and says which it is, because an assessor needs to know whether they are watching one presentation or two before they start marking.

4. Getting an account, and getting back in

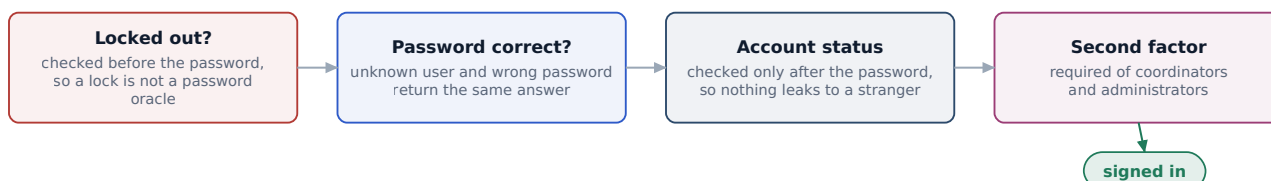
Authentication is local to the department by policy. There is no external identity provider, which makes the order of the sign-in checks and the recovery procedure part of the design rather than an implementation detail.

GETTING AN ACCOUNT

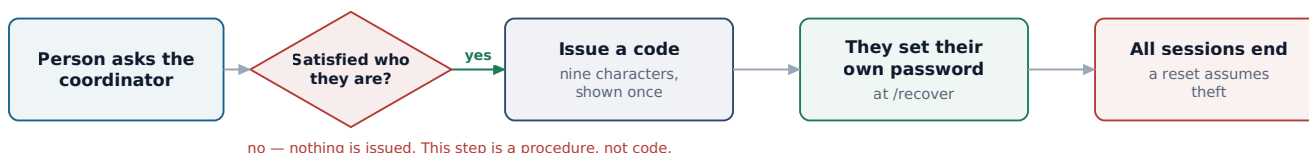


A student account is active at once. Only staff wait for approval.

SIGNING IN — THE ORDER IS THE POINT



WHEN SOMEONE CANNOT GET IN



no — nothing is issued. This step is a procedure, not code.

Figure 4. Account lifecycle and recovery. The coordinator never learns the password; the code expires in thirty minutes and works once.

The ordering is deliberate at every step. The lockout is tested before the password, so a locked account cannot be used to test passwords. Account status is tested only after the password verifies, so someone without the password learns nothing about whether an account exists or what state it is in. An unknown username and a wrong password produce the same answer.

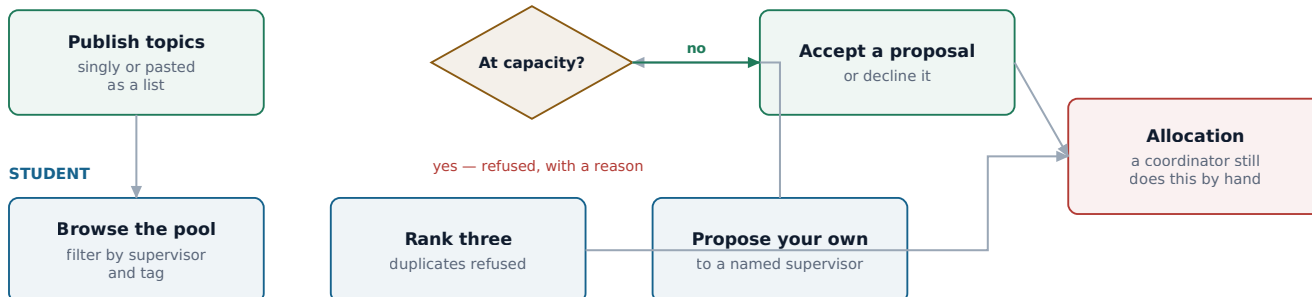
The part that is a procedure, not software

Issuing a reset code assumes the coordinator has satisfied themselves that the person asking is who they say. The system can make the code short-lived, single-use and invisible to the coordinator afterwards. It cannot recognise a voice.

5. Topics to allocation

Supervisors publish what they are willing to supervise; students express preferences; a coordinator decides.

SUPERVISOR



Ranking is not allocation, and the screen says so in as many words. Until the allocation run exists, a student's preferences are evidence for a decision that a person still makes.

Figure 5. From a topic pool to an allocated project.

Ranking three topics is not the same as being allocated one, and the screen says so in as many words, because a student who believes ranking is choosing will not chase a supervisor. A student may also propose their own topic to a named supervisor, who cannot accept it while at capacity.

Still done by hand

No screen yet turns preferences into projects. The allocation run is the largest unbuilt piece of the process, and the fairness criteria it would encode — capacity, rank satisfaction, keeping pairs together — are a departmental decision before they are a technical one.

6. Consultations

A consultation is a claim that two people met. One attestation is a note; two make a record that counts.

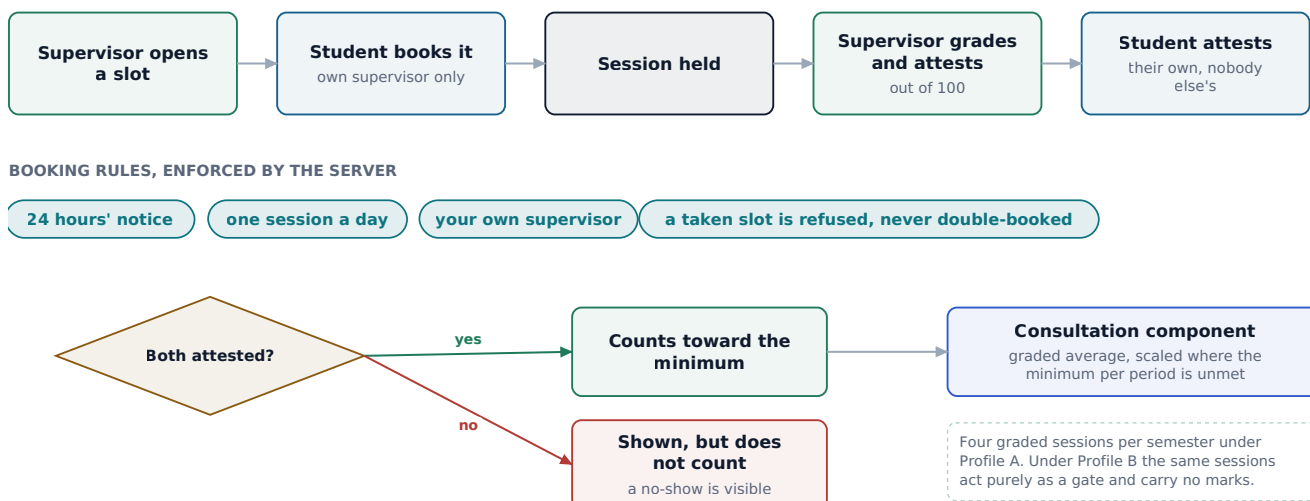


Figure 6. A consultation is a claim by two people. One attestation is a note; two make a record.

Consultations are always individual, even on a joint project. One member of a pair can meet the minimum while the other is scaled down for missing it, and the supervisor dashboard shows exactly that.

Whether consultations carry marks or act purely as a gate is the main difference between the two policy profiles, which is why the consultation rubric is still marked provisional: current practice records attendance only, so the instrument does not yet exist.

7. Marking a presentation

One sheet per assessor covering the whole cohort, mirroring the paper form: session numbers in the gutter, criteria with their maxima in the header, and totals the assessor never types.

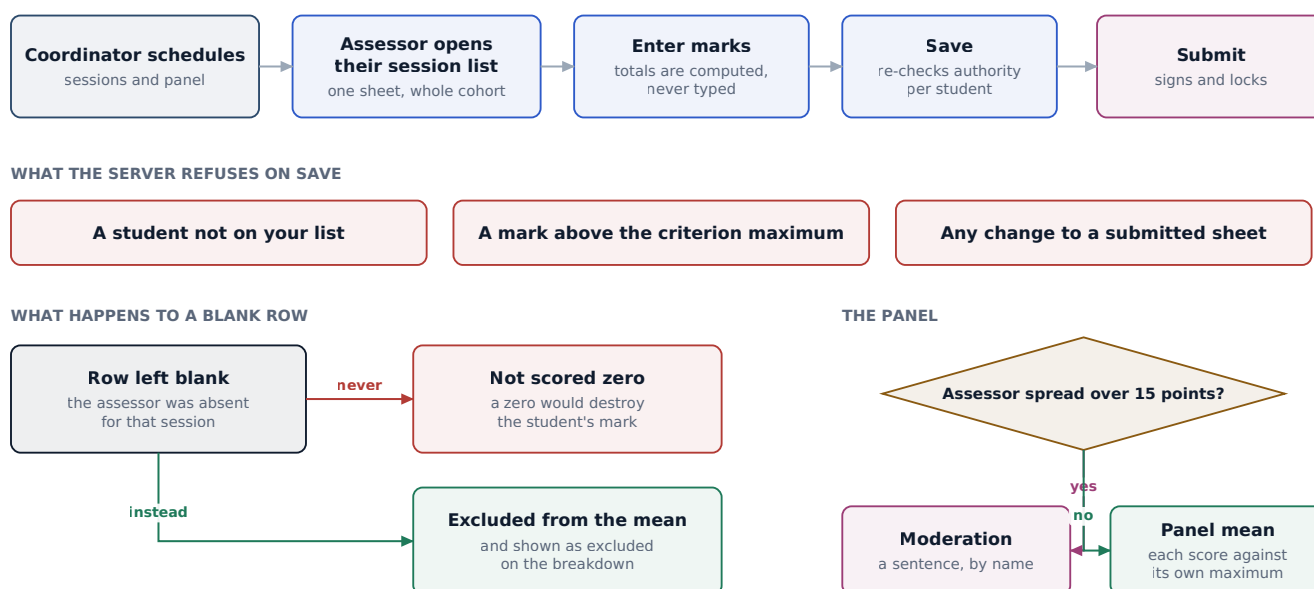


Figure 7. Marking a presentation. The three refusals and the blank-row rule are the parts that protect a mark.

The blank row is the rule most worth understanding. An assessor who leaves a student's row empty was absent for that session. Recording that as zero would destroy the student's mark while looking like a legitimate score, so the row is excluded from the panel mean instead and shown as excluded on the breakdown.

Submitting is the signature

A saved sheet can still be changed. A submitted sheet is locked against its assessor's name and the time, and only a coordinator can reopen it — and not if they produced the mark themselves.

8. How a mark is computed

Every score persists both its raw total and the maximum in force when it was awarded. Nothing downstream may use a raw total without dividing by its recorded maximum.

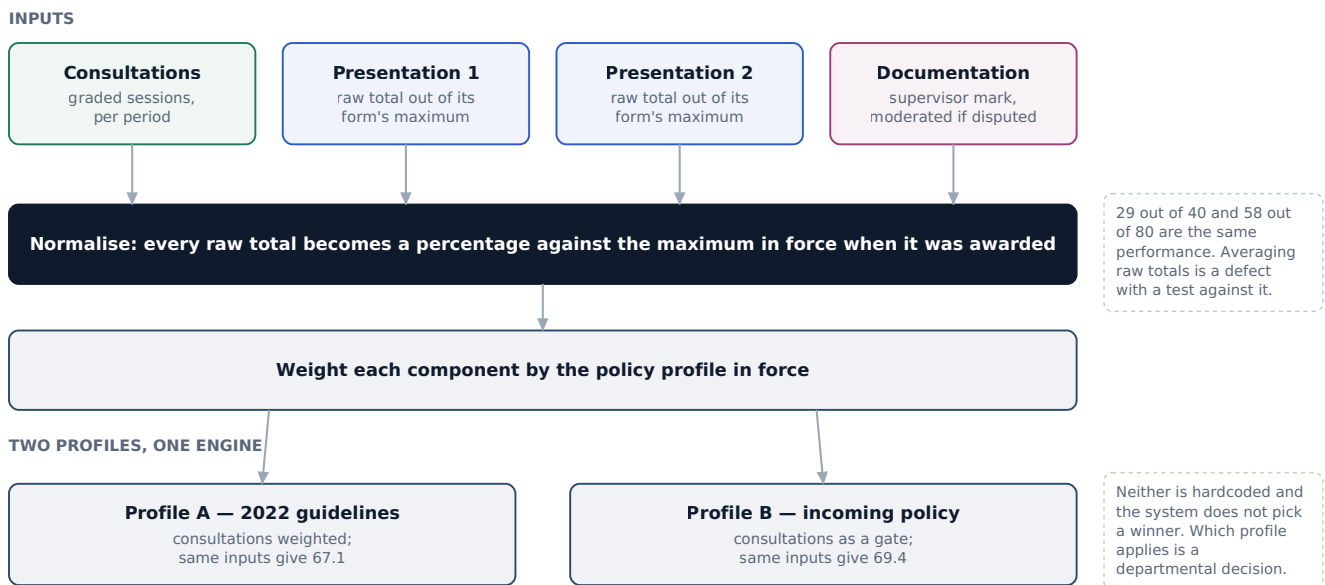


Figure 8. The mark pipeline. Normalisation happens once, in one function, and everything downstream consumes percentages.

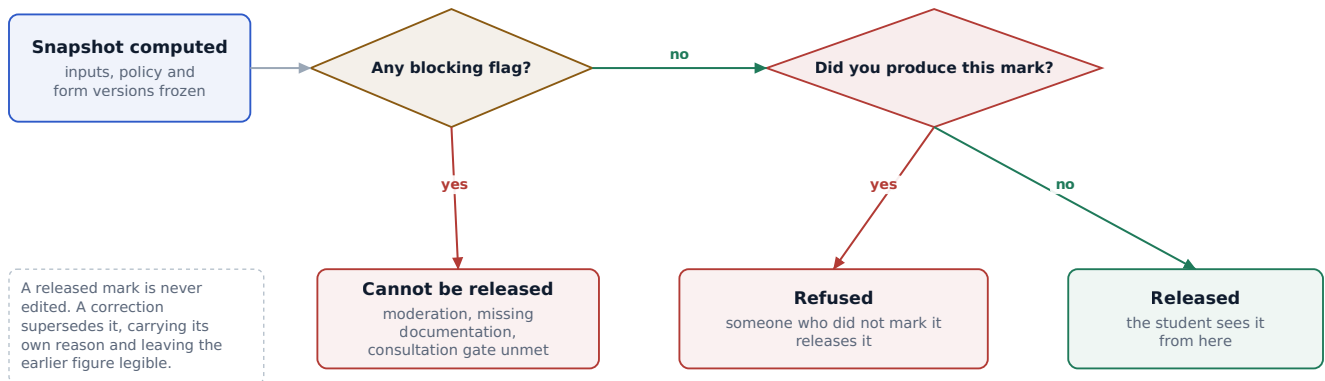
That rule is the reason the system can survive a change of assessment form, a student carrying credit from a previous cycle, and two profiles disagreeing about how the continuous assessment mark is composed. Each of those changes a maximum somewhere; none of them changes what a percentage means.

An open departmental decision

The 2022 guidelines and the incoming policy compose the continuous assessment mark differently. Both are expressed as configuration and the same inputs produce 67.1 under one and 69.4 under the other. The system deliberately does not pick.

9. Moderation and release

A mark becomes visible to a student at exactly one moment, and two gates stand in front of it.



Reports read the stored snapshot. Nothing recomputes a published mark.

Figure 9. Releasing a result. Both gates have to be passed, in this order.

A spread between assessors beyond the configured threshold does not average itself away. It blocks, and a moderator resolves it with a written rationale of at least a sentence, recorded against their name. A one-word rationale is refused.

After release, the mark is not edited. A correction supersedes it, carrying its own reason, and the earlier figure stays legible. This is the difference between a system that can defend a mark and one that can only display the current value.

10. Changing an assessment form

The columns of a marking sheet are the department's instrument, not presentation, and they change between cycles. The coordinator changes them without a code change — but never underneath a mark already awarded.

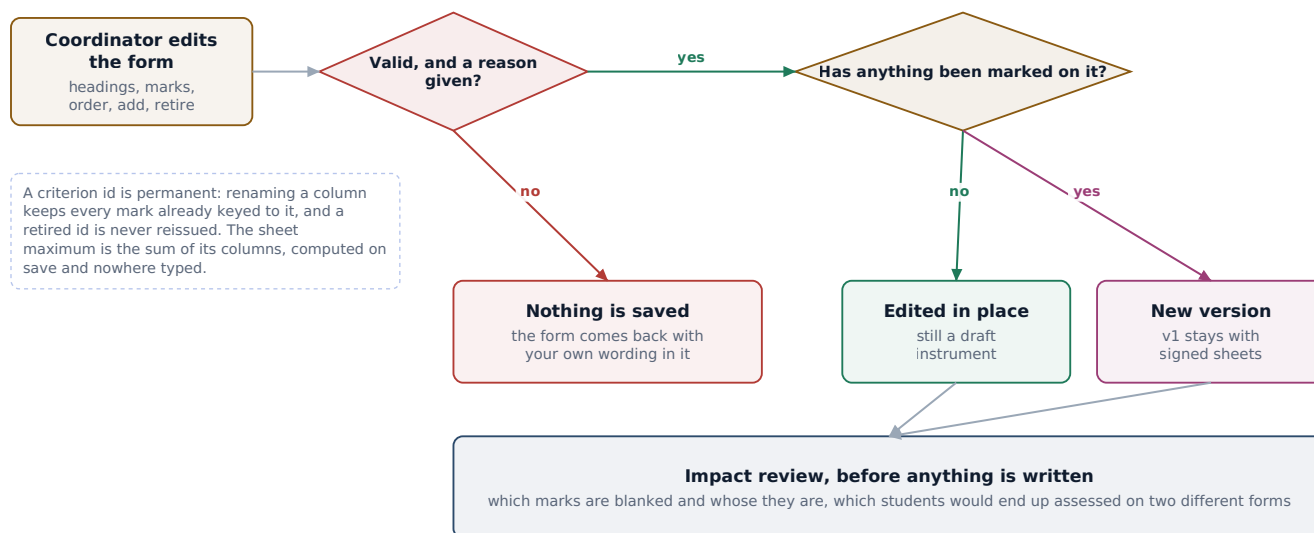


Figure 10. Editing the columns of a marking sheet. The fork is what keeps an old mark meaningful.

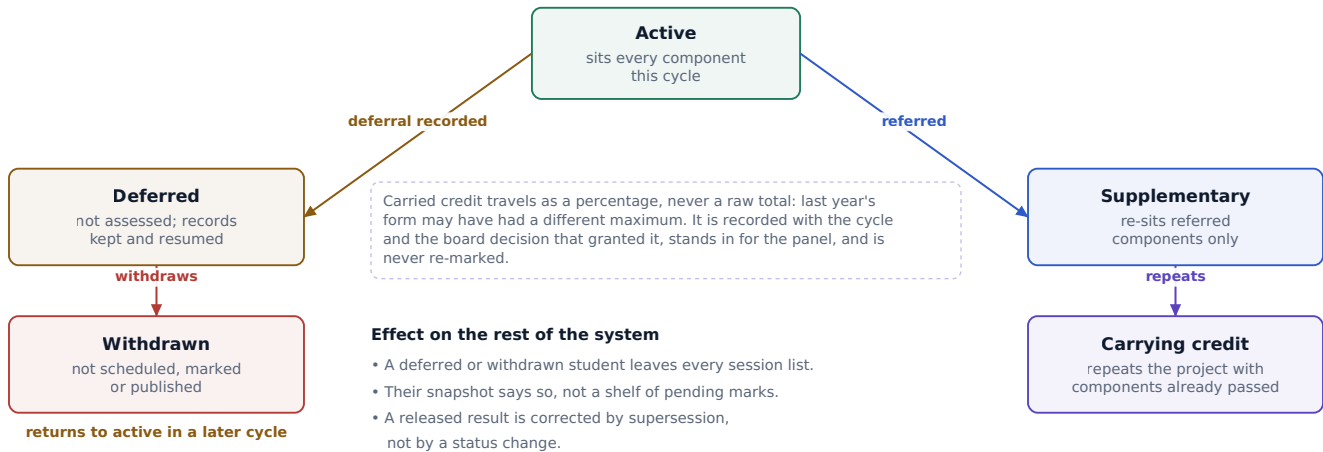
Four rules make this safe rather than merely flexible. A form nothing has been marked on is still a draft and is edited in place. The moment a mark exists, an edit publishes a new version and signed sheets stay attached to the version they were signed under. A column's identifier is permanent, so renaming a heading keeps every mark already recorded against it, and a retired identifier is never reissued. The sheet maximum is the sum of its columns, computed on save.

Lowering a column below an awarded mark

Truncating a 12 to fit a new maximum of 10 would replace an assessor's judgement with an invented one. The mark is blanked instead — excluded from the panel, not scored zero — and the assessor is asked for it again.

11. Students off the ordinary path

Real cohorts contain deferrals, withdrawals, supplementary sittings and students repeating the project while carrying a component they have already passed. Each needs an answer to two questions: are they assessed this cycle, and what is already credited.



Nothing here deletes anybody. Every record a student generated is kept.

Figure 11. The five enrolment states and the moves between them.

Nothing here deletes anybody. A withdrawn student keeps every record they generated; they simply stop being scheduled, marked and published. A status other than active requires a date it took effect — rarely the date it is recorded — and a note explaining it.

12. Deadlines, grace and extensions

Until dates live in the system, lateness is decided by whoever read the handbook last.

ONE DEADLINE, TWO STUDENTS



RULES

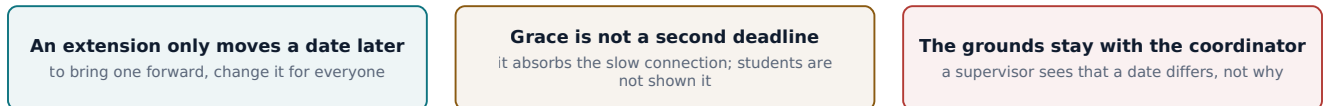


Figure 12. Deadlines, grace and extensions. An accommodation is visible as a date and private as a reason.

Grace exists for the student submitting at 23:59 on a connection that drops. It is not published, because a published grace period is simply a later deadline with an apology attached.

Accommodations

A supervisor supervising against a different date needs the date. They do not need to know it moved because of a bereavement or a disability assessment. The grounds are visible only to whoever approved them.

13. Reports and provenance

Reports read stored snapshots and never recompute. A mark schedule and the mark a student sees come from the same number, so they cannot disagree.

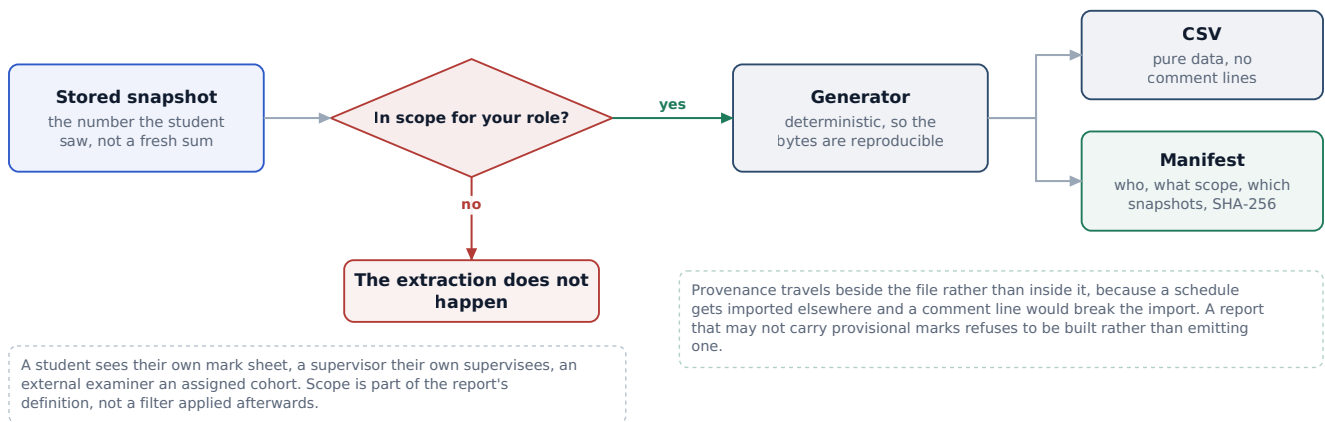


Figure 13. Producing a report. The snapshot is read, never recomputed.

Every extraction moves personal data out of the system, which under the Eswatini Data Protection Act 41 of 2022 is something a controller should be able to account for. That is why scope is part of a report's definition rather than a filter applied afterwards, and why the manifest records who pulled what, over which scope, from which snapshots, with a hash of the exact bytes issued.

14. What the system refuses

Every refusal in one place. A refusal is a policy decision made visible; if one of these is wrong for the department, it is a line of configuration or a conversation, not a workaround.

Where	What is refused	Why
Sign-in	The logout is checked before the password	A locked account must not answer whether a password was right
Sign-in	Unknown username and wrong password give the same answer	Otherwise anyone can enumerate staff and students
Marking	Posting a student who is not on your session list	Authority is re-checked per student on save, not once at page render
Marking	A mark above the criterion maximum	The criterion is named in the refusal so it can be fixed
Marking	Any change to a submitted sheet	Submitting is the signature; only a coordinator reopens it
Consultations	Attesting somebody else's session	A student attests their own attendance and nobody else's
Booking	Less than 24 hours' notice, two in a day, another supervisor	A slot already taken is refused rather than double-booked
Moderation	A one-word rationale	Moderating requires a sentence, recorded against your name
Release	Releasing a mark you produced	Separation of duty, applied per action rather than per role
Release	Releasing a blocked mark	A blocking flag has to be resolved, not overridden
Assessment forms	An edit with no reason, or fewer than two columns	The reason is kept with the version and shown to anyone auditing a mark
Enrolment	Changing the status of a student with a released result	That is a correction by supersession, not a dropdown
Extensions	An extension that moves a deadline earlier	To bring a date forward, change the deadline for everyone
Reports	A mark sheet containing an unpublished figure	The manifest refuses to build rather than emit a provisional mark

15. Not yet automated

A process handbook that describes only the built parts is a marketing document. This is what a person still does, and what happens in the meantime.

Area	Where it stands	What happens meanwhile
Allocation	Students rank and propose, supervisors accept. Nothing turns preferences into projects	A coordinator allocates by hand
Audit chain	The chain is implemented and tested; the application writes to a console sink rather than to it	Decisions are recorded in the working store only
Deliverable upload	Virus scanning is in the stack with nothing yet to scan	Files are exchanged by email
Second factor	Enforcement and enrolment exist; the verification endpoint does not	Privileged roles are challenged but cannot yet complete the challenge
Documentation marking	Specified, and the rubric is a placeholder awaiting the departmental marking guide	Marked on paper
Presentation nomination	Nomination and scheduling are specified but not built	Session lists are prepared by the coordinator
Reports	Two of fourteen generators are implemented; XLSX and PDF/A are open	The rest are produced by hand
Ethics clearance	Recorded on a project and displayed; it gates nothing	Enforced by the supervisor
Examination board	Release is a coordinator action; there is no ratification step	Ratification happens outside the system

The one to fix first

The audit chain is complete and tested, and the application does not yet write to it. Until it does, the accountability argument in section 13 describes the design rather than the running system.

None of the above affects the correctness of a mark that the system does produce. They affect how much of the process it carries, which is a different question and a more honest one to be asked about.